**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 137 · Network Threats

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. How can more active users affect network performance?

Q2. How does a brute-force attack work?
A. Tries many possible passwords
B. Sends a trusted-looking message
C. Floods a web server
D. Rearranges file fragments

Q3. “A MAC address is the text domain name entered into a browser.” Correct this. Explain why.

Q4. Data must reach another network in the right order. “TCP chooses fonts; IP draws images.” Correct this. Explain each protocol's job.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Malware — Malicious software such as ransomware, spyware or viruses.

Social engineering — Manipulating people into revealing information or bypassing safeguards.

Denial of service — Overwhelms a service to degrade or prevent access.

Unauthorised access — Guessing credentials or exploiting flaws may expose or change protected information.

## New learning

Malware: Malicious software such as ransomware, spyware or viruses.

Social engineering: Manipulating people into revealing information or bypassing safeguards.

Denial of service: Overwhelms a service to degrade or prevent access.

Unauthorised access: Guessing credentials or exploiting flaws may expose or change protected information.

## Worked example

A staff member receives a message asking for their password through an unfamiliar link. Identify the risk and response.

1. Classify the attempt as phishing/social engineering.

2. Explain that the link could harvest credentials.

3. Do not enter details; verify independently via official channels.

4. Report to school IT and follow account-security procedures.

# We Do · Practise with guidance

### Guided 1

An online service receives huge volumes of requests and slows down. Diagnose.

Worked method: Distinguish attack evidence from a normal traffic spike.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Complete the line that increments an observed failed-login count.

```
attempts = 2
...
print(attempts)
```

write

### Question 1

Give the meaning of phishing. [2 marks]

### Question 2

Explain how ransomware can harm a school. [3 marks]

### Question 3

Name one precaution against social engineering. [2 marks]

# Exit Ticket · Answer before checking

Phishing is a form of social engineering.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Users may compete for finite bandwidth, reducing the data rate available to each.

Q2. A — Tries many possible passwords

Q3. A MAC address identifies a network interface on a LAN. It is usually six pairs of hexadecimal digits.

Q4. TCP supports reliable delivery in order. IP handles addressing and routing between networks. Each protocol has its own job; encryption does not prove a site is honest.

### I Do / We Do pages · Guided practice

Q1. Distinguish attack evidence from a normal traffic spike.

### You Do pages · Code First

```
attempts = 2
attempts = attempts + 1
print(attempts)
```

Expected result: 3

### You Do pages · Independent questions

Q1. Deceptive communication designed to trick a user into disclosing data or taking unsafe actions.

Q2. Malware may encrypt/deny access to files, disrupt services and cause recovery costs.

Q3. Staff training, verifying requests independently or multifactor authentication.

### Exit Ticket · Model answer

True

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## 1. Copy the whole original question:

## 2. Write a complete new answer in GREEN PEN. Show working if needed: